SANGBAY
Gas Sensor

Laser Methane Remote Sensor

SLTD-CH4-100 Sensor Specification

Long distance | Contactless | Fast response speed | High precision

Shanghai Sangbay Sensor Technology Co., Ltd.

1 Product Overview

The SLTD-CH4-100 Laser Methane Remote Sensor (hereinafter referred to as "the sensor") is independently developed by Shanghai Sangbay Sensor Technology Co., Ltd.The sensor is based on Tunable Diode Laser Absorption Spectroscopy (TDLAS) technology, which offers significant advantages over traditional chemical and infrared spectroscopy absorption (NDIR) technologies, including:High sensitivity, capable of detecting minute leaks;High accuracy, unaffected by other gases;Easy maintenance, no need for frequent calibration;Faster response speed and longer service life;Utilizes the laser's excellent directionality and long-distance remote sensing capability to achieve active long-distance monitoring, changing the original passive monitoring method that relies on diffusion or pump suction, without waiting for leaked gas to diffuse to the monitoring equipment;Low false alarm rate and high reliability, sensitive only to methane gas, unaffected by insecticides, alcohol, paint, other alkanes, etc., suitable for environments with complex background gases.



Figure 1: SLTD-CH4-100 Laser Methane Remote Sensor Physical Image

2 Product Characteristics

- High Safety,Non-contact remote sensing with intrinsically safe design.
- Adaptable to Various Complex Conditions,Strong anti-poisoning capability and

environmental adaptability.

- Millisecond-Level Fast Response, Real-time capture of gas changes.
- Excellent Anti-Interference Capability,Unique selectivity, unaffected by other gases.
- Outstanding Stability and Low Cost,Zero-point stability without drift, no need for frequent calibration, long service life.

3 Application Field

- Gas Safety Inspection: Vehicle-mounted or handheld inspection for urban underground pipe networks, valve wells, and pressure regulation stations.
- Oil and Gas Industry Monitoring: Leak detection and carbon emission monitoring for long-distance pipelines, storage tanks, and oil/gas field facilities.
- Environmental Protection and Carbon Accounting: Precise monitoring of fugitive methane emissions at landfills, wastewater treatment plants, and other locations.
- Industrial Safety Prevention: Long-distance safety scanning in chemical parks, tunnel corridors, and hazardous areas to protect personnel and facilities.
- New Platform Integration: Mounted on drones and robots for three-dimensional intelligent inspection in complex environments (such as mountainous pipelines, factory rooftops).

4 Product Introduction

4.1 Working Principle

The laser methane remote sensor is based on Tunable Diode Laser Absorption Spectroscopy (TDLAS) technology.Its core principle utilizes the unique absorption characteristics of methane molecules at a specific wavelength of infrared laser (approximately 1653 nm). The laser inside the sensor emits a beam precisely tuned to this absorption spectral line. When the laser transmits through the air, if methane gas exists along the path, the molecules will absorb light energy, causing light intensity attenuation. The degree of attenuation is proportional to the product of methane concentration and optical path length, strictly following the Beer-Lambert physical law.

The instrument receives the reflected light from the target surface, and the detector precisely measures the light intensity changes. Using high-frequency wavelength modulation and second harmonic detection technology, it effectively overcomes background interference and finally accurately calculates the methane concentration path integral value (unit: ppm • m), achieving non-contact, long-distance, high-selectivity rapid detection of methane leaks within distances of tens to hundreds of meters.

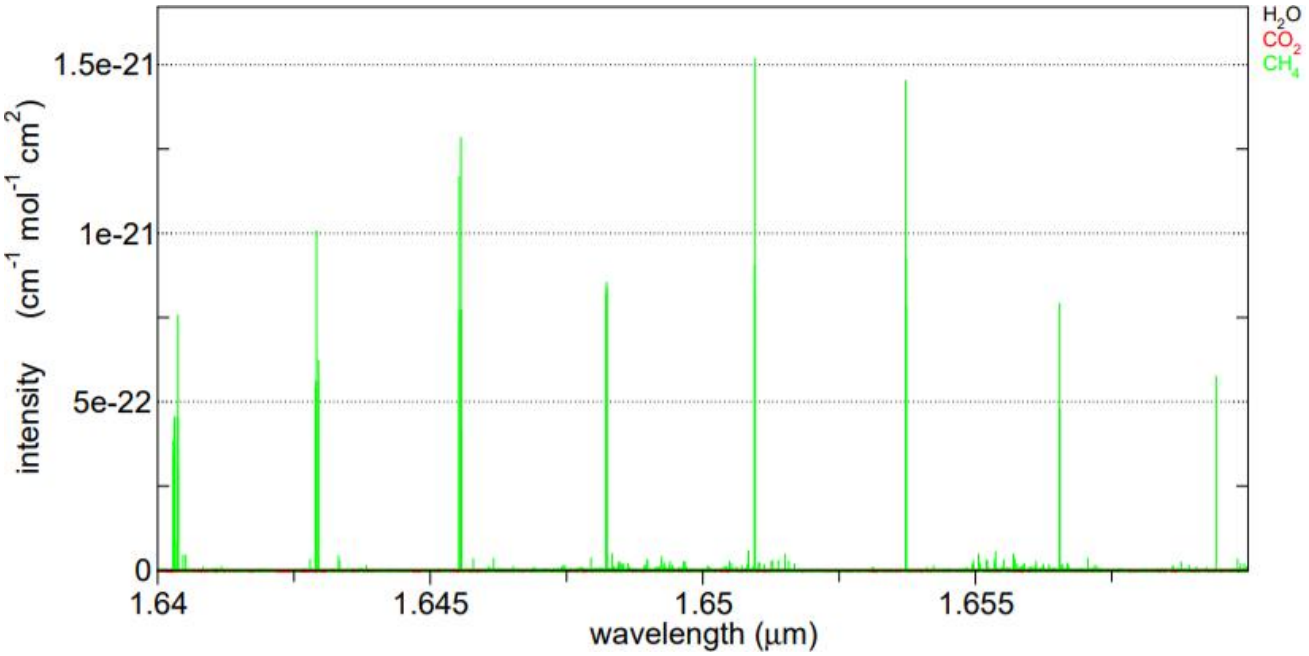


Figure 2: Methane Absorption Spectral Line Near 1653 nm

In remote sensing detection, the emitted beam must pass through the leaked methane gas cloud, be reflected by a reflective surface, pass through the gas cloud again, and return to the detector for the instrument to complete the measurement and output the unit concentration value. Since the leaked methane gas cloud drifts with the wind, it is constantly changing and unevenly distributed in both space and time; the distance from the instrument to the reflector is uncertain, and the thickness of the methane gas cloud is also uncertain. To express the concentration value under such circumstances, the industry commonly uses the concept of unit concentration (ppm • m), which represents: Methane Concentration (ppm) × Standard Thickness (1m).

4.2 Unit Concentration (ppm • m)

"100 ppm • m means: 100 ppm of methane gas uniformly distributed over a 1-meter thick gas cloud".It can be approximately considered that unit concentration is the concentration after uniformly "compressing" or "expanding" the methane molecules on the straight line (laser detection line) between the remote sensor and the reflective target into a 1-meter thick region.

5 Dimensions and Interface Definition

5.1 Sensor Dimensions

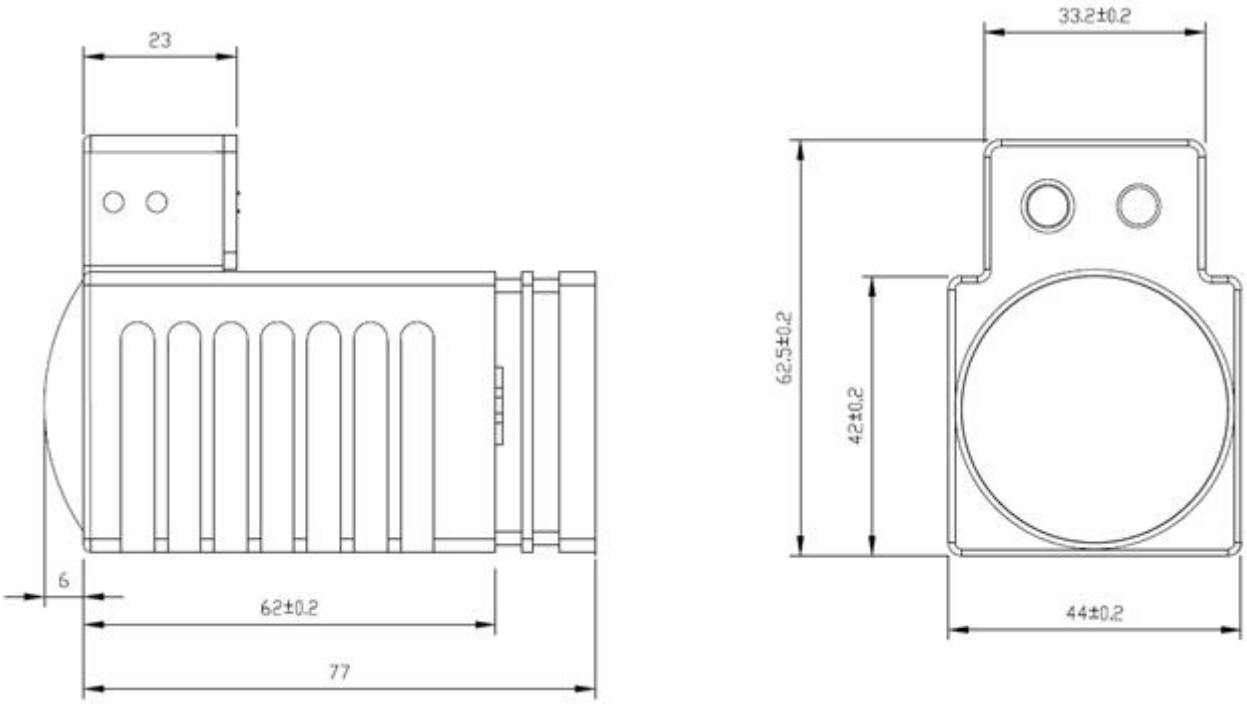


Figure 3: 100-Meter Laser Methane Remote Sensor Dimension Diagram

Dimension unit: mm

5.2 Interface Definition

The sensor is equipped with 5 interfaces. The specific pin definitions are shown below:

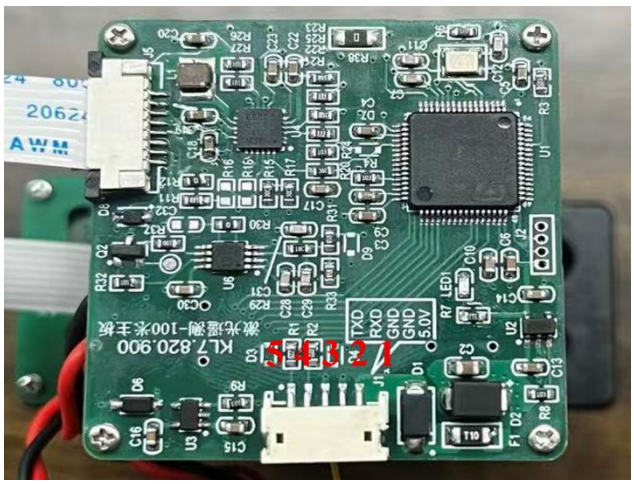


Figure 4:Sensor Pin Definition Diagram

Table 1: 100-Meter Laser Methane Remote Sensor Pin Definition

Pin	Definition
1	DC+
2	GND
3	GND
4	RX
5	TX

6 Technical Specifications

Table 2: 100-Meter Laser Methane Remote Sensor Technical Specifications

Target Gas	Methane (CH4)
Detection Principle	Tunable Laser Spectroscopy Absorption Technology (TDLAS)
Detection Distance	100 meters
Detection Range	0-100000ppm.m
Basic Error	±10%

Sensitivity	5ppm.m
Response Time	≤0.05s
Signal Output	TTL Serial Communication (see attachment for communication protocol)
Voltage	DC 5V
Power Consumption	< 2W
Operating Temperature	− 20 ~ 50℃
Operating Humidity	≤98%RH, non-condensing
Material	Aluminum Alloy
Dimensions	82 × 44 × 62.6 mm(L × W × H)
Application Scenarios	Gas inspection, petrochemical, environmental monitoring, gas power generation, landfill, etc.

7 Transmission Protocol

7.1 Physical Layer Communication Protocol

UART Communication Level Standard: TTL（5V） Communication Interface UART3.3

UART Parameter Configuration:

Baud Rate	115200b/s
Data Bits	8
Stop Bits	1
Parity Bit	**None**

7.2 Protocol Communication Format

The module's communication protocol is divided into two types: Data Output and Module Control.

- Module Control Frame: Send control commands and wait for acknowledgment.
- Data Output Frame: Send data externally at a fixed 50ms cycle.

(1) Data Output Frame Format

Byte	1	2	.....	2+N	3+N	4+N	5+N
Message （HEX）	Frame Header	Data LengthN	.....	Data N	Checksum Low	Checksum High	Frame Tail

Data Length: Number of bytes occupied by data, N=1,2,3……

(2) Module Control Frame Format

Send Control Frame:

Byte	1	2	3	4	5	6	7	8	9
Message （HEX）	Frame Header	Length	Control Command	Control ValueH	Control ValueL	Data Items ...	Checksum Low	Checksum High	Frame Tail

Control Frame Acknowledgment:

Byte	1	2	3	4	5	6	7	8	9
Message （HEX）	Frame Header	Length	Control Command	Control ValueH	Control ValueL	Data Items ...	Checksum Low	Checksum High	Frame Tail

- Control Code + Control Value + Data Items
- Checksum: CRC16 (Length + Control Command + Control Value + Data Items)

7.3 Communication Content

Send Control Frame:

Byte	1	2	3	4	5	6	7	8	9

Message（HEX）	AA	Length	Control Code	Control ValueH	Control ValueL	Data Items...	Checksum Low	Checksum High	55

Control Frame Acknowledgment:

Byte	1	2	3	4	5	6	7	8	9
Message **（HEX）**	BB	Length	Control Code	Control ValueH	Control ValueL	Execution Result	Checksum Low	Checksum High	66

Error Type: 0x00 represents no error type

Execution Result: 0x00 represents success, 0x01 represents failure

Control Code Description:

Control Command	Control code	Control Value		Data Items	Data Description
Zero Calibration	0x09	00	00	None	Data zero calibration
Set Temperature Compensation Coefficient	0x0A	00	00	Temperature（float，low byte first）Temperature Compensation Coefficient（float，low byte first）	Set high/low temperature compensation coefficients
Measurement ON	0x10	00	00	None	None Start dynamic measurement, green light OFF
		01	00	None	None Start dynamic measurement, green light ON
		00	01		None Start static

				None	measurement, green light OFF
		01	01	None	None Start static measurement, green light ON
Measurement OFF	0x11	00	00	None	None Stop measurement
Calibration	0x12	00	00	Temperature Value (2-byte Hex, low byte first)	Calibration

Data Output:

Byte	Message（**HEX**）
1	CC
2	Length （0x09）
3	0x01
4	Concentration Value Low（HEX）
5	Concentration Value
6	Concentration Value
7	Concentration Value High
8	Light Intensity Value Low(HEX)
9	Light Intensity Value High
10	Laser Real-time Temperature Low (HEX)
11	Laser Real-time Temperature High
12	Checksum Low
13	Checksum High
14	77

Concentration Value Range: 0~100000

Concentration Value, Light Intensity Value, and Temperature Value data types are integers.

After the device receives the Measurement ON command, it starts outputting data. After receiving the Measurement OFF command, it stops outputting data.

When testing high-concentration gas, if the concentration value displays as 100000.

8 Precautions

- When the device is powered on, do not look directly at or use optical instruments (such as telescopes or magnifying glasses) to observe the green indicator laser and infrared laser at the front of the instrument, to avoid eye injury!
- When the device is powered on, do not point the green indicator laser at the faces of other people or animals, as this action may cause accidental injury or damage!
- Do not attempt to repair or replace instrument components yourself! When the instrument cannot work normally or displays error messages, please refer to the relevant descriptions in this manual for repair operations, or contact the manufacturer's technical personnel.
- Avoid prolonged direct sunlight exposure to the device, or placing it in a vehicle under direct sunlight, and using the device in adverse weather conditions such as rain, snow, hail, or strong winds! It is recommended to store the device in a special packaging box when not in use.